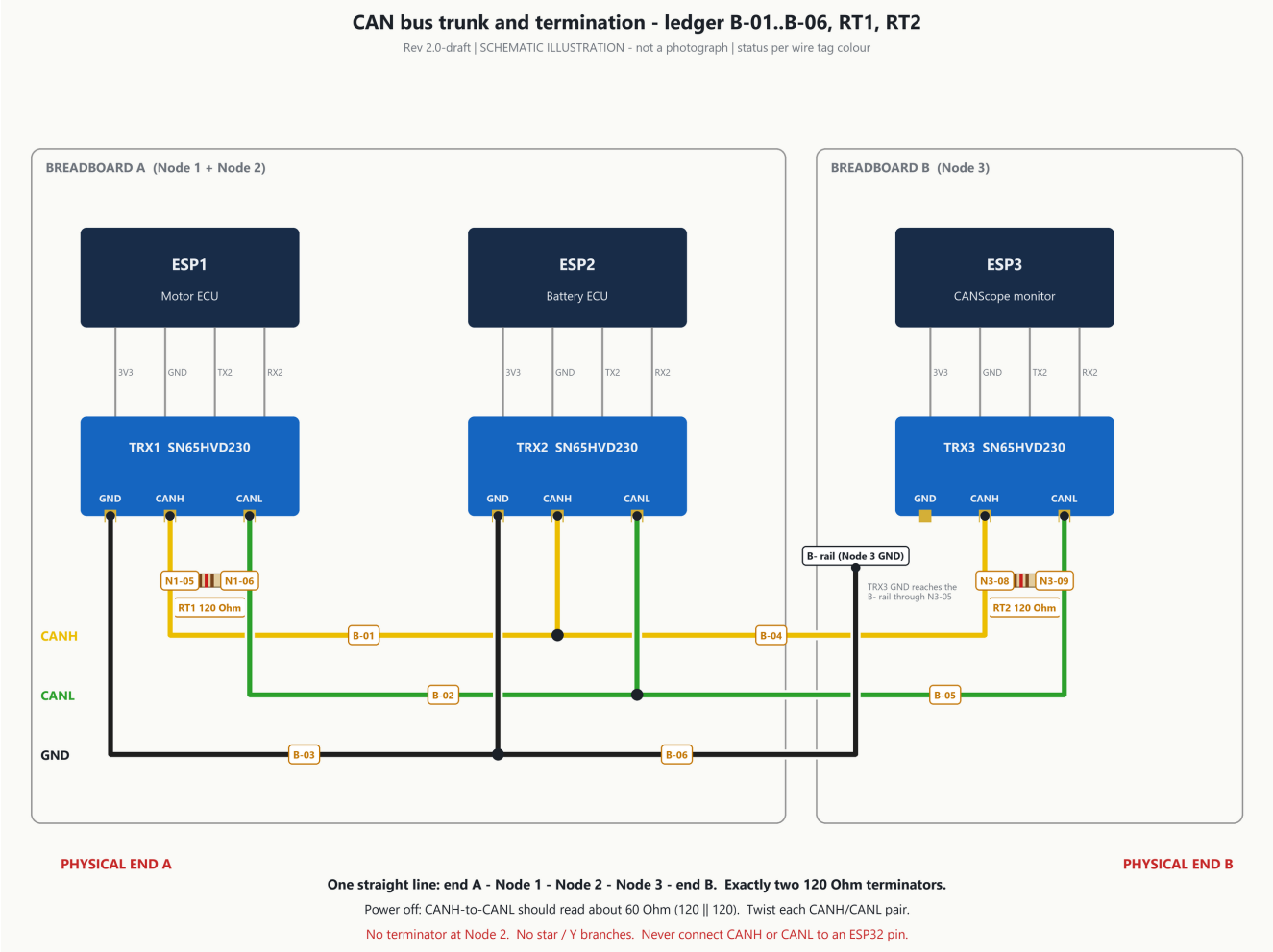
CANScope

Three-node ESP32 + Classical CAN bench: physical build manual

Rev 2.0-draft | 2026-10-16 | supersedes Rev 1.0 | generated from `docs/wiring/pin_ledger.json`



NOT FINAL. This revision corrects Rev 1.0 and assigns every GPIO, but no board, module or wire has been physically inspected, measured or powered by its author. Every diagram is a schematic illustration, not a photograph of your parts. PROVISIONAL connections may be made only after the bench gates listed for them pass. BLOCKED connections (the green LED, N3-14) must stay empty. A successful compile never proves the hardware works.

0. How to use this manual

Every wire has one row in the canonical ledger (`docs/wiring/pin_ledger.json`). The diagrams, the video, this PDF and the sign-off checklist are generated from it, and `tools/hwdocs/validate.py` checks that they agree with each other and with the firmware pin settings. If anything disagrees with the ledger, the ledger wins. Report the mismatch.

Status	Meaning	What you do
CONFIRMED	Supported by a named document and independent of unidentified parts	Wire it; check labels
PROVISIONAL	Supported by documents but depends on a detail only you can see	Pass the listed gates first
BLOCKED	Required information is missing	Leave it EMPTY; supply the evidence

Gates G1-G14 are short bench checks. Record each result on the sign-off sheet (section 18).

Corrections to Rev 1.0

The full register, with evidence, is in `validation/correction_register.md` (27 entries). The most important:

- Node 3 CAN moves to GPIO17/16 (the same as Nodes 1/2 and the repo's monitor firmware). The brief's 22/21 collided with the Arduino-ESP32 default I2C pins SDA 21 / SCL 22, which the OLED needs.

- The SN65HVD230 has no STB/EN pin. Pin 8 is Rs, which must be low for high-speed mode; high means standby (cannot transmit or ACK). Vref stays open.

- Node 3 peripheral GPIOs: OLED 21/22, LEDs 26/27; the green LED has no verified pin (N3-14 BLOCKED). None are strapping, flash, input-only or USB-serial pins.

- Push-buttons and the buzzer are removed from this build: no wiring, no firmware.

- Onboard terminators: measured per module (G6) with a decision rule; exactly two on the bus.

- The 3V3 of different nodes is never joined; Board A rails carry GND only.

- Bare boards are flashed before wiring; the bring-up firmware for each node is named.

- Firmware gaps stated plainly: no OLED/LED code, no simulation, no fault injection yet.

1. Inventory and tools

Qty	Item	Status	Notes
3	ESP32 DevKit V1, 30-pin, ESP-WROOM-32	available	G1 shield marking, G2 header order
3	SN65HVD230 CAN transceiver module	available	G4 identity, G5 Rs pin, G6 onboard terminator
1	SSD1306 0.96 in I2C OLED 128x64, 4-pin	available	G7 label order and address
1 each	Green, yellow, red LED	available	G9 polarity
3	220 Ohm resistor (one per LED)	available	G13 measure
2	120 Ohm resistor (bus ends)	available	G13; fit only per G6
2	830-point breadboard	available	G14 rails
many	Dupont jumper wires	available	Colours suggested in the ledger
3	USB data cable	available	Charge-only cables will not flash
1	Digital multimeter (continuity, diode, Ohm, DC V)	REQUIRED TOOL	Every gate uses it

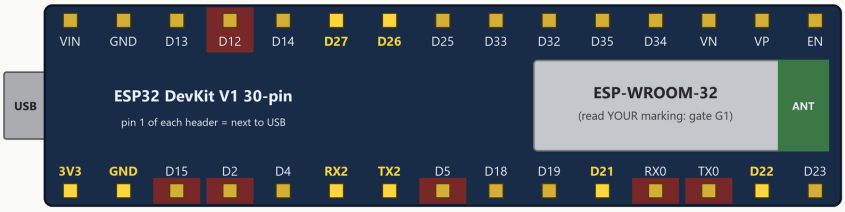
2. Identification guide

ESP32 DevKit V1 (30-pin) pin locator: USB to the LEFT, component side up

Highlighted = used by CANScope. Dark red = do NOT use (flash, strapping, USB serial).

VIN-side header (pin 1 = next to USB)

1	VIN	power	
2	GND	power	
3	D13	GPIO13	
4	D12	GPIO12	
5	D14	GPIO14	
6	**D27**	**GPIO27**	**used 3:R_R**
7	**D26**	**GPIO26**	**used 3:R_Y**
8	D25	GPIO25	
9	D33	GPIO33	
10	D32	GPIO32	
11	D35	GPIO35	
12	D34	GPIO34	
13	VN	GPIO39	
14	VP	GPIO36	
15	EN	power	

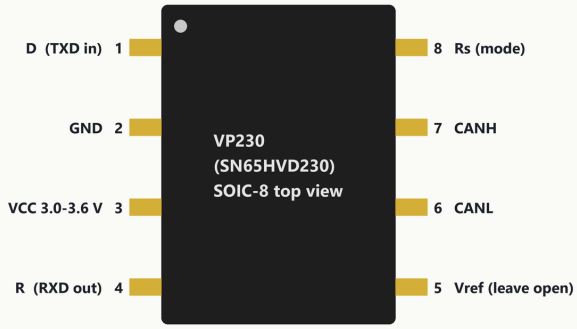


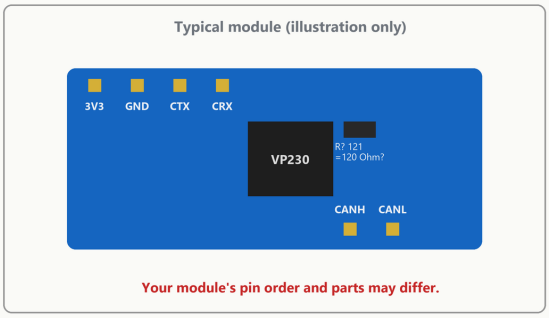
3V3-side header (pin 1 = next to USB)

1	**3V3**	**power**	**used 1:TRX1, 2:TRX2, 3:RAIL3**
2	**GND**	**power**	**used 1:TRX1, 2:TRX2, 3:RAIL3**
3	D15	GPIO15	
4	D2	GPIO2	
5	D4	GPIO4	
6	**RX2**	**GPIO16**	**used 1:TRX1, 2:TRX2, 3:TRX3**
7	**TX2**	**GPIO17**	**used 1:TRX1, 2:TRX2, 3:TRX3**
8	D5	GPIO5	
9	D18	GPIO18	
10	D19	GPIO19	
11	**D21**	**GPIO21**	**used 3:OLED**
12	RX0	GPIO3	
13	TX0	GPIO1	
14	**D22**	**GPIO22**	**used 3:OLED**
15	D23	GPIO23	

Gate G2: your silkscreen must match this order exactly, or STOP.

SN65HVD230 transceiver: what the IC is, and what to check on YOUR module



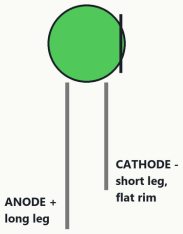


Supply	VCC 3.0-3.6 V. Feed it from the ESP32 3V3 pin. Never 5 V (D/R abs max = VCC+0.5 V).
TX path	ESP32 TX GPIO -> module CTX / TXD / D. NOT crossed like a UART.
RX path	Module CRX / RXD / R -> ESP32 RX GPIO.
Rs pin 8	Strong pull-down = high-speed (needed). 10-100 kOhm = slope control. High = STANDBY: driver off, cannot transmit or ACK. Gate G5.
Vref pin 5	VCC/2 reference output. Leave unconnected.
Termination	Many modules carry a 120 Ohm across CANH-CANL. Measure it (gate G6).
Labels	Module header order varies. Wire by PRINTED NAME only (gate G4).

Source: Texas Instruments SN65HVD23x datasheet SLOS346O (rev. April 2018), pin table and Section 8.

Identify before you insert: LED, resistor, OLED

LED polarity (gate G9)



GPIO -> 220 Ohm -> anode; cathode -> GND

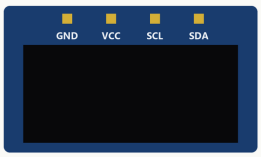
Resistor colour bands (4-band) - measure anyway (G13)

220 Ohm: red red brown (gold)

120 Ohm: brown red brown (gold)

5-band (1%) parts use different colours.

SSD1306 I2C OLED (G7)

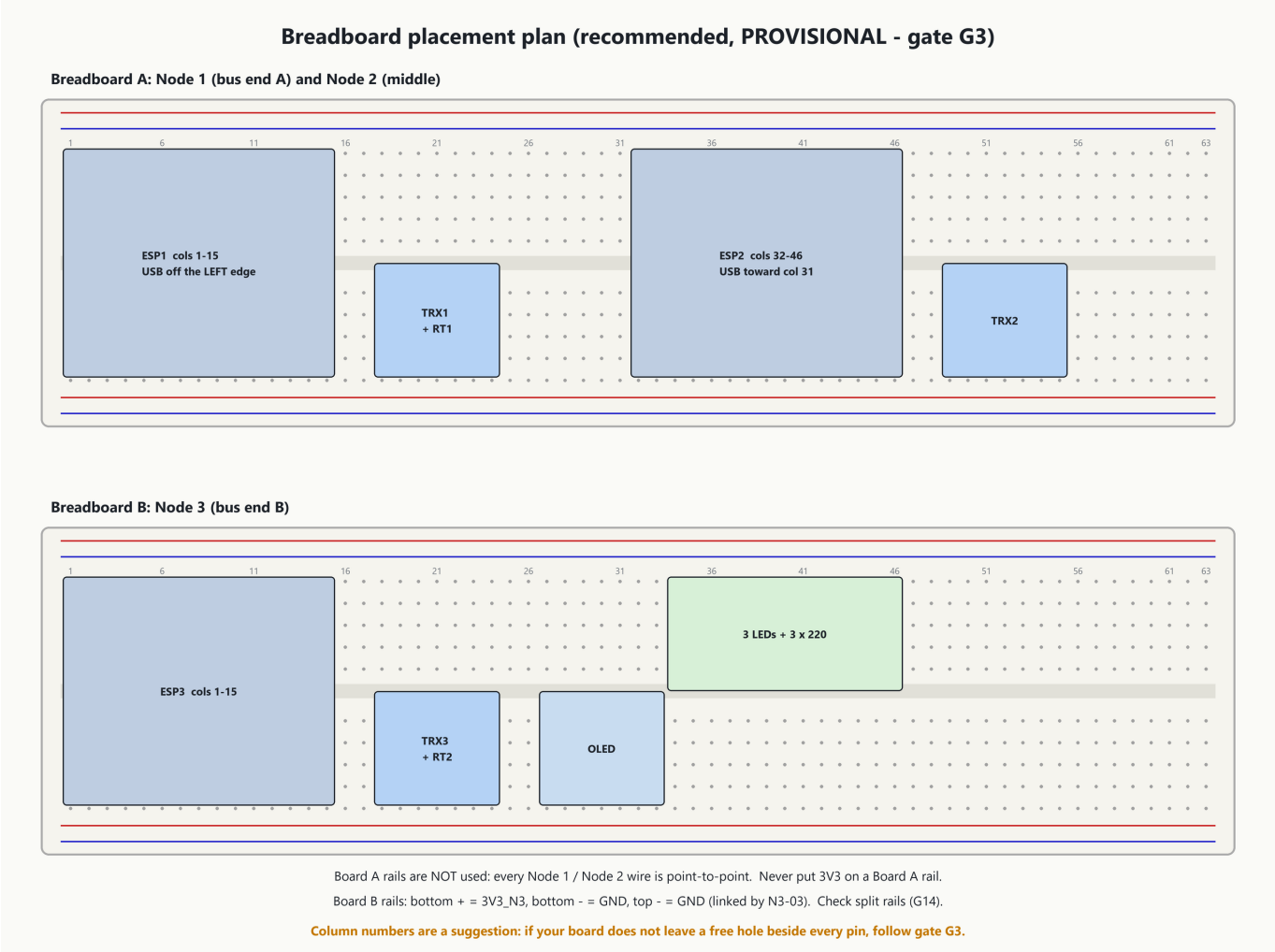


Read YOUR labels: some modules swap GND and VCC.
Power from 3V3 only (keeps SDA/SCL pull-ups at 3.3 V).
Address: back marking 0x78 = 0x3C.

Bench gates

Gate	Title	When	Procedure	Pass	Fail
G1	ESP32 module identity	Before any wire	Read the laser marking on the metal shield of each ESP32. Photograph it.	It reads ESP32-WROOM-32 (or WROOM-32D/32E/32U). Not WROVER, not ESP32-S2/S3/C3.	STOP. GPIO16/17 may be PSRAM pins. Send the photo; the ledger must be revised.
G2	Header silkscreen matches the ledger	Before any wire	With USB pointing left, read every pin label on both headers and compare with the 30-pin table (3V3 side: 3V3, GND, D15, D2, D4, RX2, TX2, D5, D18, D19, D21, RX0, TX0, D22, D23; VIN side: VIN, GND, D13, D12, D14, D27, D26, D25, D33, D32, D35, D34, VN, VP, EN).	All 30 labels match in the same order.	STOP. Do not count pins from the diagram. Send a photo of both headers.
G3	Breadboard fit	When seating each ESP32	Seat the ESP32 across the centre trench. Count free holes beside each pin row.	At least one free hole remains in the same 5-hole strip as every pin, on both sides.	Do not force it. Bridge the board across two breadboards (one header row in each) or use female-to-male jumpers on the header pins. Record which.
G4	Transceiver module identity and labels	Before wiring a transceiver	Read the 8-pin IC marking (expect VP230) and every printed pin label. Check whether the module has any 5 V pin or a voltage regulator. Photograph front and back.	IC reads VP230; supply pin is labelled 3V3/3.3V/VCC with no 5 V-only marking; you can find pins equivalent to 3V3, GND, CTX (TXD/D), CRX (RXD/R), CANH, CANL.	STOP. Different IC or a 5 V module needs a new design review.
G5	Transceiver Rs (mode) pin	After G4	If the module exposes a pin labelled RS, S or STB: it must be wired to GND (see the conditional rows). If not exposed: with all power removed, measure resistance from IC pin 8 (Rs) to IC pin 2 (GND).	Exposed RS pin wired to GND, OR measured Rs-GND below 100 Ohm (high-speed mode per TI SLOS346O).	Rs-GND 10-100 kOhm (slope control) or open/high (standby, cannot transmit or ACK): STOP and record the value.
G6	Onboard termination	Before installing RT1/RT2	With the module unpowered and NOT connected to anything, measure CANH to CANL on each transceiver module.	Reading above 10 kOhm (no onboard terminator) on all three modules: fit RT1 and RT2 exactly as the ledger says.	About 120 Ohm on a module: it has an onboard terminator. On TRX1/TRX3 omit the external RT1/RT2 at that node. On TRX2 (middle node): STOP, the onboard resistor must be removed or its jumper opened; send a photo first.
G7	OLED labels	Before wiring the OLED	Read the four header labels left to right and the address marking on the back (0x78 = 0x3C, 0x7A = 0x3D).	Labels GND, VCC (or VDD), SCL (or SCK), SDA are present; record their order and address.	Different labels (e.g. SPI pins CS/DC/RES): STOP, it is not an I2C module.
G9	LED polarity	Before inserting each LED	Long leg = anode (+). Flat edge on the rim = cathode (-). Confirm with the meter diode-test mode (red probe on anode lights the LED faintly).	Anode and cathode identified for every LED.	Legs cut equal and no flat edge visible: use diode test only.
G13	Resistor values	Before inserting each resistor	Measure every resistor out of circuit.	220 Ohm: 209-231; 120 Ohm: 114-126; 1 kOhm: 950-1050; 10 kOhm: 9.5k-10.5k.	Wrong value: set it aside.
G14	Breadboard rails	Before using a rail	Continuity from one end of each rail to the other.	Continuous end to end, or the split point is marked.	Split rail: add a short bridge jumper on the same rail only, or keep loads on one half.

3. Breadboard layout and node labelling



Label each board with tape before placing: N1 Motor ECU, N2 Battery ECU, N3 CANScope. Board A holds Nodes 1 and 2; Board B holds Node 3. The column numbers are suggestions; gate G3 decides the fit.

Before any wire: flash each **bare** board (nothing connected but USB), one at a time. Node 1: `pio run -d firmware/node1 -e node1 -t upload`. Node 2: `pio run -d firmware/node2 -e legacy_sketch -t upload`. Node 3: `pio run -d firmware/monitor -e monitor -t upload`. All use CAN TX GPIO17, RX GPIO16, 500 kbit/s, which matches the ledger (checked by validate.py).

4. Pin-by-pin wiring ledger (all connections)

ESP32 positions count from the USB end of each header, as in the pin locator. 'Lead' means a component's own leg sharing a breadboard strip; no jumper. Colours are suggestions; the ID label on the wire is authoritative.

ID	From	To	Signal	Wire	Status	Gates	Check (power off)
N1-01	ESP1 3V3, 3V3-side pin 1 from USB	TRX1 3V3	3V3_N1	red	PROVISIONAL	G1, G2, G4	Power off: continuity ESP1 3V3 pin to TRX1 3V3 pin; no continuity to any GND or 5 V point.
N1-02	ESP1 GND, 3V3-side pin 2 from USB	TRX1 GND	GND	black	PROVISIONAL	G2, G4	Continuity ESP1 GND to TRX1 GND.
N1-03	ESP1 TX2 (GPIO17), 3V3-side pin 7 from USB	TRX1 CTX	CAN_TX_N1 (GPIO17)	white	PROVISIONAL	G1, G2, G4	Continuity TX2 pin to CTX pin; NOT to CRX.
N1-04	ESP1 RX2 (GPIO16), 3V3-side pin 6 from USB	TRX1 CRX	CAN_RX_N1 (GPIO16)	brown	PROVISIONAL	G1, G2, G4	Continuity RX2 pin to CRX pin; NOT to CTX.
N1-05	RT1 1	TRX1 CANH	CANH (end A terminator)	lead	PROVISIONAL	G6, G13	Fit only if G6 found no onboard terminator on TRX1.
N1-06	RT1 2	TRX1 CANL	CANL (end A terminator)	lead	PROVISIONAL	G6, G13	RT1 measured 114-126 Ohm before fitting.
N2-01	ESP2 3V3, 3V3-side pin 1 from USB	TRX2 3V3	3V3_N2	red	PROVISIONAL	G1, G2, G4	Continuity ESP2 3V3 to TRX2 3V3. NO continuity to Node 1 3V3.
N2-02	ESP2 GND, 3V3-side pin 2 from USB	TRX2 GND	GND	black	PROVISIONAL	G2, G4	Continuity ESP2 GND to TRX2 GND.
N2-03	ESP2 TX2 (GPIO17), 3V3-side pin 7 from USB	TRX2 CTX	CAN_TX_N2 (GPIO17)	white	PROVISIONAL	G1, G2, G4	Continuity TX2 pin to CTX pin; NOT to CRX.
N2-04	ESP2 RX2 (GPIO16), 3V3-side pin 6 from USB	TRX2 CRX	CAN_RX_N2 (GPIO16)	brown	PROVISIONAL	G1, G2, G4	Continuity RX2 pin to CRX pin; NOT to CTX.
N3-01	ESP3 3V3, 3V3-side pin 1 from USB	Board B bottom + rail (3V3_N3)	3V3_N3	red	PROVISIONAL	G2, G14	Continuity 3V3 pin to bottom + rail.
N3-02	ESP3 GND, 3V3-side pin 2 from USB	Board B bottom - rail (GND)	GND	black	PROVISIONAL	G2, G14	Continuity GND pin (3V3 side) to bottom - rail.
N3-03	Board B bottom - rail (GND)	Board B top - rail (GND)	GND (rail link, far right end of board B)	black	PROVISIONAL	G14	Continuity bottom - rail to top - rail: near 0 Ohm. Never link a - rail to a + rail.
N3-04	Board B bottom + rail (3V3_N3)	TRX3 3V3	3V3_N3	red	PROVISIONAL	G4	Continuity + rail to TRX3 3V3.
N3-05	Board B bottom - rail (GND)	TRX3 GND	GND	black	PROVISIONAL	G4	Continuity - rail to TRX3 GND.
N3-06	ESP3 TX2 (GPIO17), 3V3-side pin 7 from USB	TRX3 CTX	CAN_TX_N3 (GPIO17)	white	PROVISIONAL	G1, G2, G4	Continuity TX2 to CTX; NOT to CRX.
N3-07	ESP3 RX2 (GPIO16), 3V3-side pin 6 from USB	TRX3 CRX	CAN_RX_N3 (GPIO16)	brown	PROVISIONAL	G1, G2, G4	Continuity RX2 to CRX; NOT to CTX.
N3-08	RT2 1	TRX3 CANH	CANH (end B terminator)	lead	PROVISIONAL	G6, G13	Fit only if G6 found no onboard terminator on TRX3.
N3-09	RT2 2	TRX3 CANL	CANL (end B terminator)	lead	PROVISIONAL	G6, G13	RT2 measured 114-126 Ohm before fitting.
N3-10	Board B bottom + rail (3V3_N3)	OLED VCC	3V3_N3	red	PROVISIONAL	G7	Continuity + rail to the pin PRINTED VCC/VDD (not by position).
N3-11	Board B bottom - rail (GND)	OLED GND	GND	black	PROVISIONAL	G7	Continuity - rail to the pin PRINTED GND.
N3-12	ESP3 D21 (GPIO21), 3V3-side pin 11 from USB	OLED SDA	I2C_SDA (GPIO21)	blue	PROVISIONAL	G2, G7	Continuity D21 to SDA; NOT to SCL.
N3-13	ESP3 D22 (GPIO22), 3V3-side pin 14 from USB	OLED SCL	I2C_SCL (GPIO22)	purple	PROVISIONAL	G2, G7	Continuity D22 to SCL; NOT to SDA.

ID	From	To	Signal	Wire	Status	Gates	Check (power off)
N3-14	ESP3 D25 (GPIO25), VIN-side pin 8 from USB	R_G 1	LED_GREEN (GPIO25)	grey	BLOCKED	G2, G13	Do NOT wire. The green LED has no verified pin (GPIO25 was proposed but not confirmed). Firmware leaves the green LED disabled until a verified GPIO is recorded here.
N3-15	R_G 2	LED_G A	LED_GREEN via 220	lead	PROVISIONAL	G9	Resistor and anode share one 5-hole strip; nothing else in it.
N3-16	LED_G K	Board B top - rail (GND)	GND	lead	PROVISIONAL	G9, G14	Cathode (short leg, flat side) to top GND rail.
N3-17	ESP3 D26 (GPIO26), VIN-side pin 7 from USB	R_Y 1	LED_YELLOW (GPIO26)	grey	PROVISIONAL	G2, G13	Continuity D26 to one end of the yellow 220 Ohm.
N3-18	R_Y 2	LED_Y A	LED_YELLOW via 220	lead	PROVISIONAL	G9	Resistor and anode share one strip.
N3-19	LED_Y K	Board B top - rail (GND)	GND	lead	PROVISIONAL	G9, G14	Cathode to top GND rail.
N3-20	ESP3 D27 (GPIO27), VIN-side pin 6 from USB	R_R 1	LED_RED (GPIO27)	grey	PROVISIONAL	G2, G13	Continuity D27 to one end of the red 220 Ohm.
N3-21	R_R 2	LED_R A	LED_RED via 220	lead	PROVISIONAL	G9	Resistor and anode share one strip.
N3-22	LED_R K	Board B top - rail (GND)	GND	lead	PROVISIONAL	G9, G14	Cathode to top GND rail.
B-01	TRX1 CANH	TRX2 CANH	CANH trunk, segment 1	yellow	PROVISIONAL	G4	Continuity TRX1 CANH to TRX2 CANH; no continuity CANH to CANL other than through terminators.
B-02	TRX1 CANL	TRX2 CANL	CANL trunk, segment 1	green	PROVISIONAL	G4	Twist with B-01. Continuity TRX1 CANL to TRX2 CANL.
B-03	TRX1 GND	TRX2 GND	GND reference, segment 1	black	PROVISIONAL	G4	Continuity Node 1 GND to Node 2 GND.
B-04	TRX2 CANH	TRX3 CANH	CANH trunk, segment 2 (board A to B)	yellow	PROVISIONAL	G4	Continuity TRX2 CANH to TRX3 CANH.
B-05	TRX2 CANL	TRX3 CANL	CANL trunk, segment 2 (board A to B)	green	PROVISIONAL	G4	Twist with B-04. Continuity TRX2 CANL to TRX3 CANL.
B-06	TRX2 GND	Board B bottom - rail (GND)	GND reference, segment 2	black	PROVISIONAL	G4, G14	Continuity Node 2 GND to Node 3 GND rail.

Conditional rows

ID	Applies when	From	To	Status	Gates
C-RS1	Only if TRX1 exposes a pin labelled RS, S or STB	TRX1:RS	TRX1:GND	PROVISIONAL	G4, G5
C-RS2	Only if TRX2 exposes a pin labelled RS, S or STB	TRX2:RS	TRX2:GND	PROVISIONAL	G4, G5
C-RS3	Only if TRX3 exposes a pin labelled RS, S or STB	TRX3:RS	RAIL3:GND-	PROVISIONAL	G4, G5
C-VREF	Any module exposing Vref	TRXn:Vref	(no connection)	CONFIRMED	
C-T2	If G6 finds an onboard 120 Ohm on TRX2 (middle node)	TRX2 onboard terminator	(remove / open jumper)	BLOCKED	G6

5. Node 1 - Motor ECU wiring

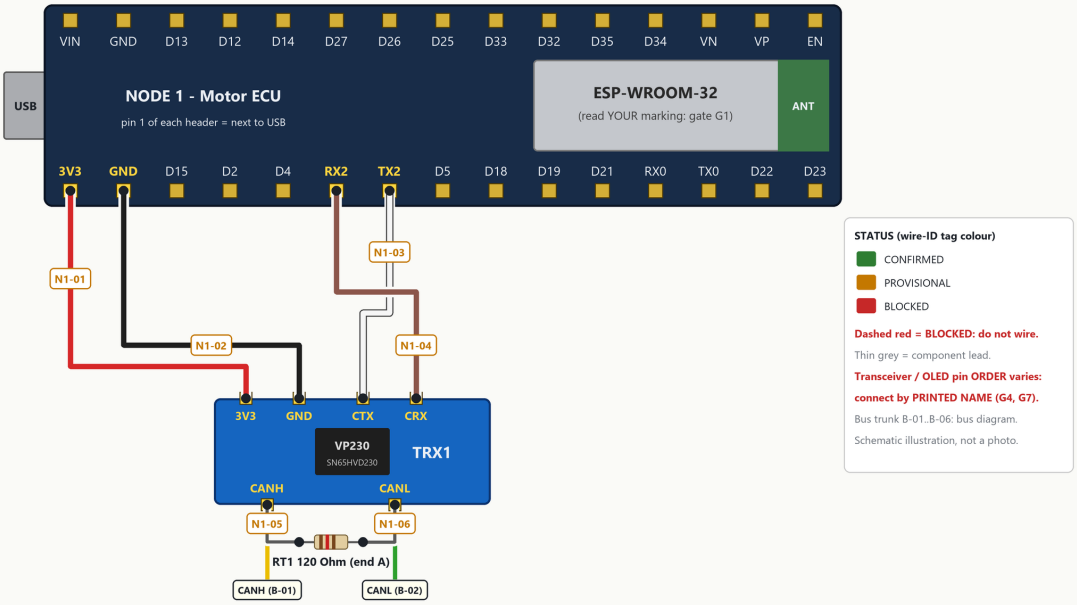
Node 1 (Motor ECU) wiring - ledger N1-01..N1-06

Rev 2.0-draft | SCHEMATIC ILLUSTRATION - not a photograph | status per wire tag colour

Node 1 uses only FOUR ESP32 pins, all on the 3V3-side header (bottom row):

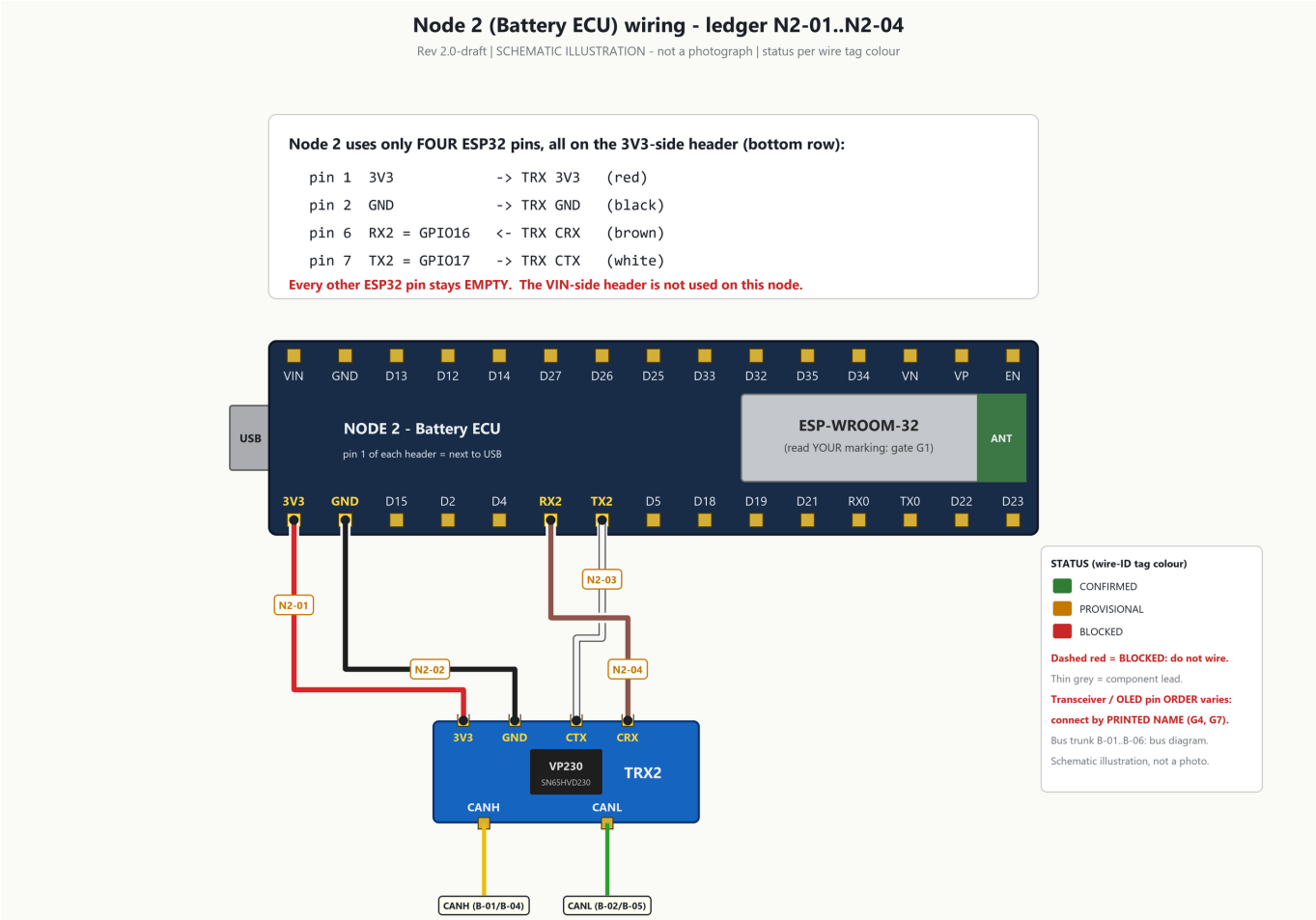
```
pin 1  3V3            -> TRX 3V3   (red)
pin 2  GND            -> TRX GND   (black)
pin 6  RX2 = GPIO16   <- TRX CRX   (brown)
pin 7  TX2 = GPIO17   -> TRX CTX   (white)
```

Every other ESP32 pin stays EMPTY. The VIN-side header is not used on this node.



ID	From	To	Signal	Wire	Status	Gates	Check (power off)
N1-01	ESP1 3V3, 3V3-side pin 1 from USB	TRX1 3V3	3V3_N1	red	**PROVISIONAL**	G1, G2, G4	Power off: continuity ESP1 3V3 pin to TRX1 3V3 pin; no continuity to any GND or 5 V point.
N1-02	ESP1 GND, 3V3-side pin 2 from USB	TRX1 GND	GND	black	**PROVISIONAL**	G2, G4	Continuity ESP1 GND to TRX1 GND.
N1-03	ESP1 TX2 (GPIO17), 3V3-side pin 7 from USB	TRX1 CTX	CAN_TX_N1 (GPIO17)	white	**PROVISIONAL**	G1, G2, G4	Continuity TX2 pin to CTX pin; NOT to CRX.
N1-04	ESP1 RX2 (GPIO16), 3V3-side pin 6 from USB	TRX1 CRX	CAN_RX_N1 (GPIO16)	brown	**PROVISIONAL**	G1, G2, G4	Continuity RX2 pin to CRX pin; NOT to CTX.
N1-05	RT1 1	TRX1 CANH	CANH (end A terminator)	lead	**PROVISIONAL**	G6, G13	Fit only if G6 found no onboard terminator on TRX1.
N1-06	RT1 2	TRX1 CANL	CANL (end A terminator)	lead	**PROVISIONAL**	G6, G13	RT1 measured 114-126 Ohm before fitting.

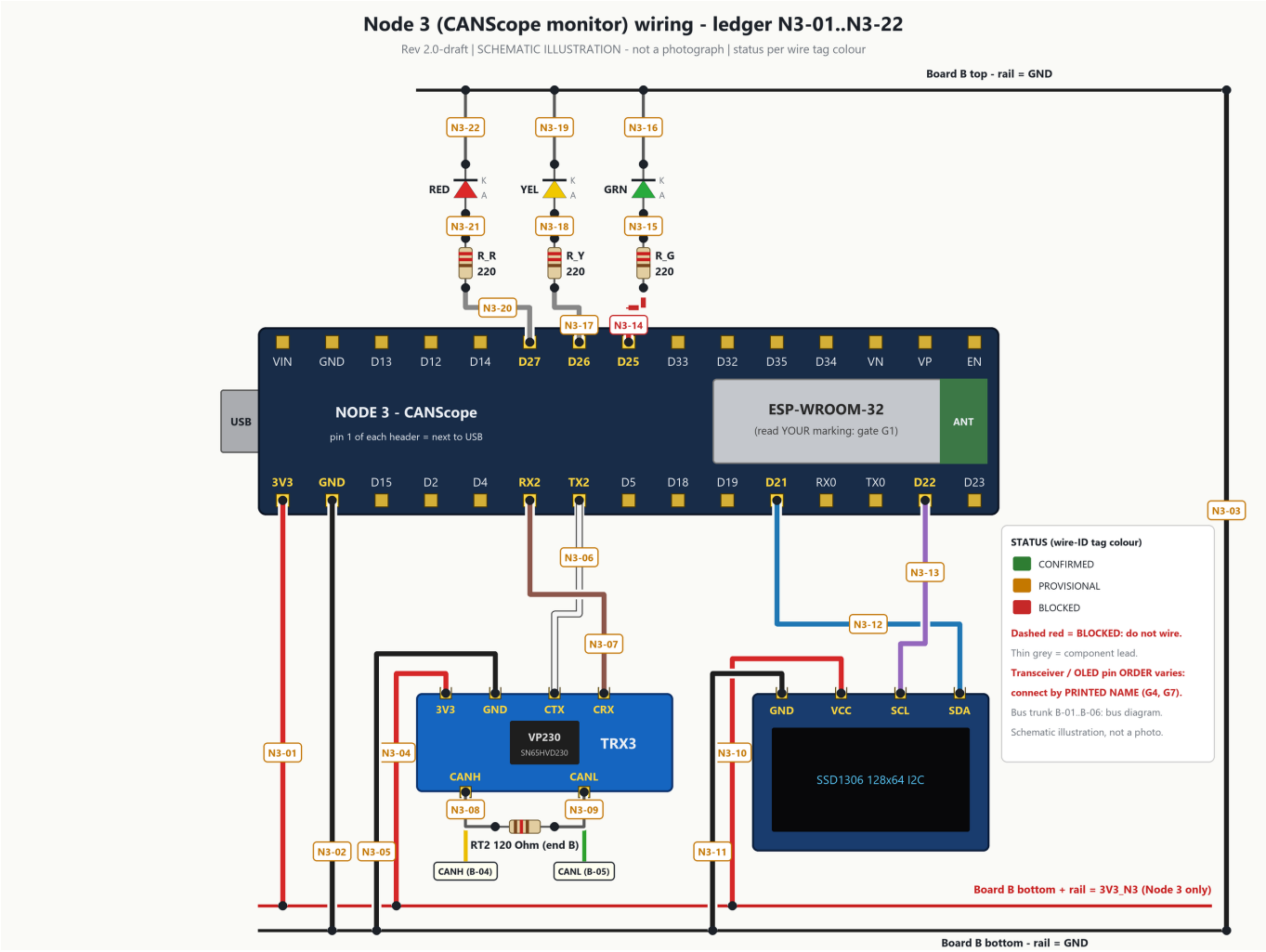
6. Node 2 - Battery ECU wiring



ID	From	To	Signal	Wire	Status	Gates	Check (power off)
N2-01	ESP2 3V3, 3V3-side pin 1 from USB	TRX2 3V3	3V3_N2	red	**PROVISIONAL**	G1, G2, G4	Continuity ESP2 3V3 to TRX2 3V3. NO continuity to Node 1 3V3.
N2-02	ESP2 GND, 3V3-side pin 2 from USB	TRX2 GND	GND	black	**PROVISIONAL**	G2, G4	Continuity ESP2 GND to TRX2 GND.
N2-03	ESP2 TX2 (GPIO17), 3V3-side pin 7 from USB	TRX2 CTX	CAN_TX_N2 (GPIO17)	white	**PROVISIONAL**	G1, G2, G4	Continuity TX2 pin to CTX pin; NOT to CRX.
N2-04	ESP2 RX2 (GPIO16), 3V3-side pin 6 from USB	TRX2 CRX	CAN_RX_N2 (GPIO16)	brown	**PROVISIONAL**	G1, G2, G4	Continuity RX2 pin to CRX pin; NOT to CTX.

Shared with Node 1: the same four pins and colours. Node-specific: Node 2 is the middle of the bus (no terminator), and its 3V3 must never touch Node 1's 3V3.

7. Node 3 - CANScope monitor wiring



Rows for Node 3 are listed by circuit in sections 8-9; power and transceiver rows below.

ID	From	To	Signal	Wire	Status	Gates	Check (power off)
N3-01	ESP3 3V3, 3V3-side pin 1 from USB	Board B bottom + rail (3V3_N3)	3V3_N3	red	PROVISIONAL	G2, G14	Continuity 3V3 pin to bottom + rail.
N3-02	ESP3 GND, 3V3-side pin 2 from USB	Board B bottom - rail (GND)	GND	black	PROVISIONAL	G2, G14	Continuity GND pin (3V3 side) to bottom - rail.
N3-03	Board B bottom - rail (GND)	Board B top - rail (GND)	GND (rail link, far right end of board B)	black	PROVISIONAL	G14	Continuity bottom - rail to top - rail: near 0 Ohm. Never link a - rail to a + rail.
N3-04	Board B bottom + rail (3V3_N3)	TRX3 3V3	3V3_N3	red	PROVISIONAL	G4	Continuity + rail to TRX3 3V3.
N3-05	Board B bottom - rail (GND)	TRX3 GND	GND	black	PROVISIONAL	G4	Continuity - rail to TRX3 GND.
N3-06	ESP3 TX2 (GPIO17), 3V3-side pin 7 from USB	TRX3 CTX	CAN_TX_N3 (GPIO17)	white	PROVISIONAL	G1, G2, G4	Continuity TX2 to CTX; NOT to CRX.
N3-07	ESP3 RX2 (GPIO16), 3V3-side pin 6 from USB	TRX3 CRX	CAN_RX_N3 (GPIO16)	brown	PROVISIONAL	G1, G2, G4	Continuity RX2 to CRX; NOT to CTX.

8. OLED wiring and orientation

Wire by the printed pin name, never by position: modules exist with GND-VCC-SCL-SDA and VCC-GND-SCL-SDA orders (gate G7). Power from the 3V3 rail only, so the module's I2C pull-ups sit at 3.3 V. SDA = GPIO21 and SCL = GPIO22 are the Arduino-ESP32 default Wire pins (pins_arduino.h). The address is usually 0x3C (back marking 0x78). It can only be confirmed by an I2C scan, and **no firmware for it exists yet**.

ID	From	To	Signal	Wire	Status	Gates	Check (power off)
N3-10	Board B bottom + rail (3V3_N3)	OLED VCC	3V3_N3	red	PROVISIONAL	G7	Continuity + rail to the pin PRINTED VCC/VDD (not by position).
N3-11	Board B bottom - rail (GND)	OLED GND	GND	black	PROVISIONAL	G7	Continuity - rail to the pin PRINTED GND.
N3-12	ESP3 D21 (GPIO21), 3V3-side pin 11 from USB	OLED SDA	I2C_SDA (GPIO21)	blue	PROVISIONAL	G2, G7	Continuity D21 to SDA; NOT to SCL.
N3-13	ESP3 D22 (GPIO22), 3V3-side pin 14 from USB	OLED SCL	I2C_SCL (GPIO22)	purple	PROVISIONAL	G2, G7	Continuity D22 to SCL; NOT to SDA.

9. Three LED circuits

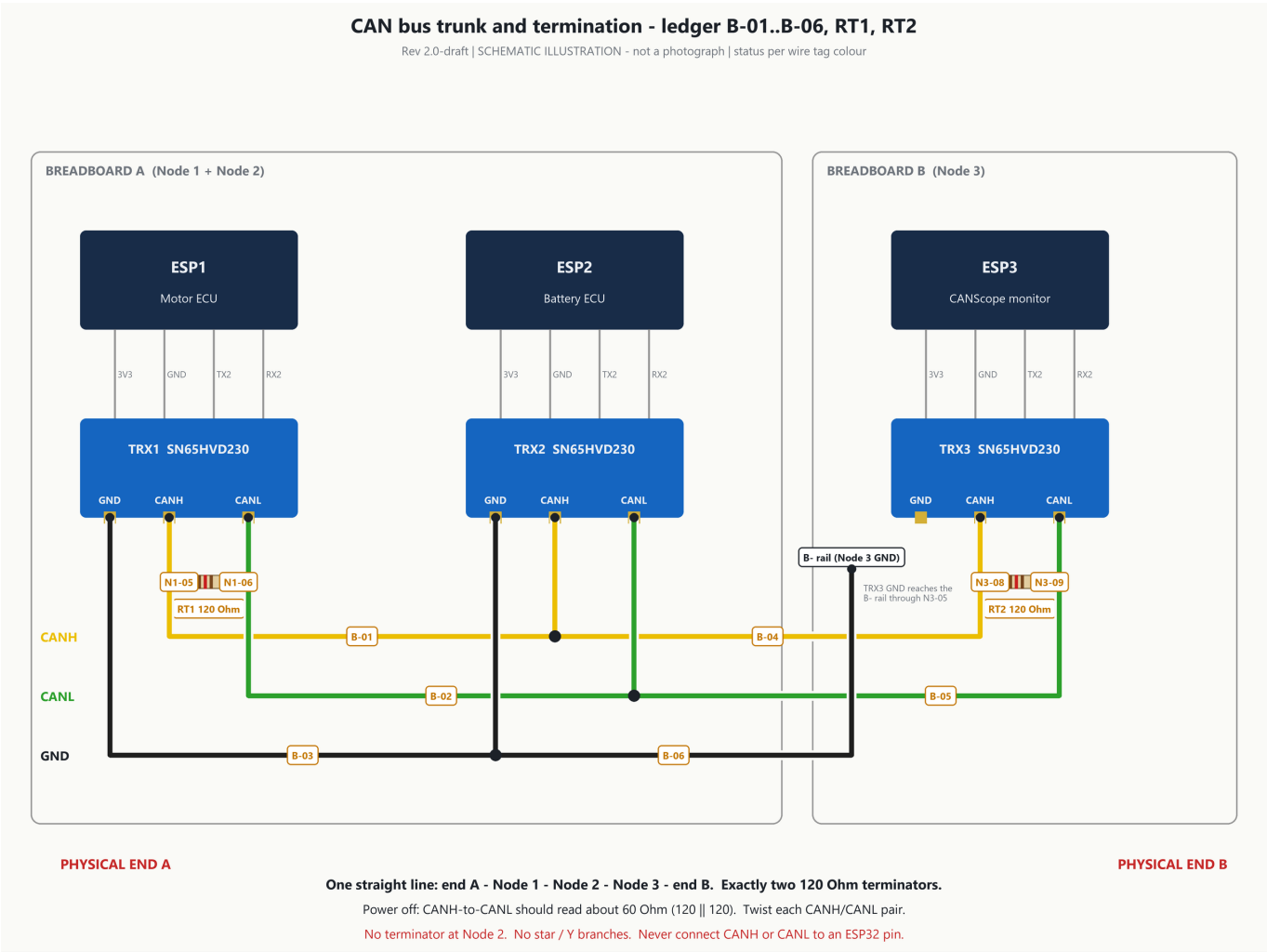
Each LED has its own 220 Ohm series resistor: GPIO -> resistor -> anode (long leg); cathode (short leg, flat rim) -> GND. Current = (3.3 V - Vf) / 220 Ohm, about 5-7 mA. Never share one resistor between LEDs.

ID	From	To	Signal	Wire	Status	Gates	Check (power off)
N3-14	ESP3 D25 (GPIO25), VIN-side pin 8 from USB	R_G 1	LED_GREEN (GPIO25)	grey	BLOCKED	G2, G13	Do NOT wire. The green LED has no verified pin (GPIO25 was proposed but not confirmed). Firmware leaves the green LED disabled until a verified GPIO is recorded here.
N3-15	R_G 2	LED_G A	LED_GREEN via 220	lead	PROVISIONAL	G9	Resistor and anode share one 5-hole strip; nothing else in it.
N3-16	LED_G K	Board B top - rail (GND)	GND	lead	PROVISIONAL	G9, G14	Cathode (short leg, flat side) to top GND rail.
N3-17	ESP3 D26 (GPIO26), VIN-side pin 7 from USB	R_Y 1	LED_YELLOW (GPIO26)	grey	PROVISIONAL	G2, G13	Continuity D26 to one end of the yellow 220 Ohm.
N3-18	R_Y 2	LED_Y A	LED_YELLOW via 220	lead	PROVISIONAL	G9	Resistor and anode share one strip.
N3-19	LED_Y K	Board B top - rail (GND)	GND	lead	PROVISIONAL	G9, G14	Cathode to top GND rail.
N3-20	ESP3 D27 (GPIO27), VIN-side pin 6 from USB	R_R 1	LED_RED (GPIO27)	grey	PROVISIONAL	G2, G13	Continuity D27 to one end of the red 220 Ohm.
N3-21	R_R 2	LED_R A	LED_RED via 220	lead	PROVISIONAL	G9	Resistor and anode share one strip.
N3-22	LED_R K	Board B top - rail (GND)	GND	lead	PROVISIONAL	G9, G14	Cathode to top GND rail.

10. Push-buttons and buzzer: removed

Node 3 has no push-buttons and no buzzer: they were removed from this build, and nothing is wired to GPIO25, GPIO32, GPIO33 or VIN. The OLED turns its pages by itself.

11. Physical CANH/CANL trunk



ID	From	To	Signal	Wire	Status	Gates	Check (power off)
B-01	TRX1 CANH	TRX2 CANH	CANH trunk, segment 1	yellow	PROVISIONAL	G4	Continuity TRX1 CANH to TRX2 CANH; no continuity CANH to CANL other than through terminators.
B-02	TRX1 CANL	TRX2 CANL	CANL trunk, segment 1	green	PROVISIONAL	G4	Twist with B-01. Continuity TRX1 CANL to TRX2 CANL.
B-03	TRX1 GND	TRX2 GND	GND reference, segment 1	black	PROVISIONAL	G4	Continuity Node 1 GND to Node 2 GND.
B-04	TRX2 CANH	TRX3 CANH	CANH trunk, segment 2 (board A to B)	yellow	PROVISIONAL	G4	Continuity TRX2 CANH to TRX3 CANH.
B-05	TRX2 CANL	TRX3 CANL	CANL trunk, segment 2 (board A to B)	green	PROVISIONAL	G4	Twist with B-04. Continuity TRX2 CANL to TRX3 CANL.
B-06	TRX2 GND	Board B bottom - rail (GND)	GND reference, segment 2	black	PROVISIONAL	G4, G14	Continuity Node 2 GND to Node 3 GND rail.

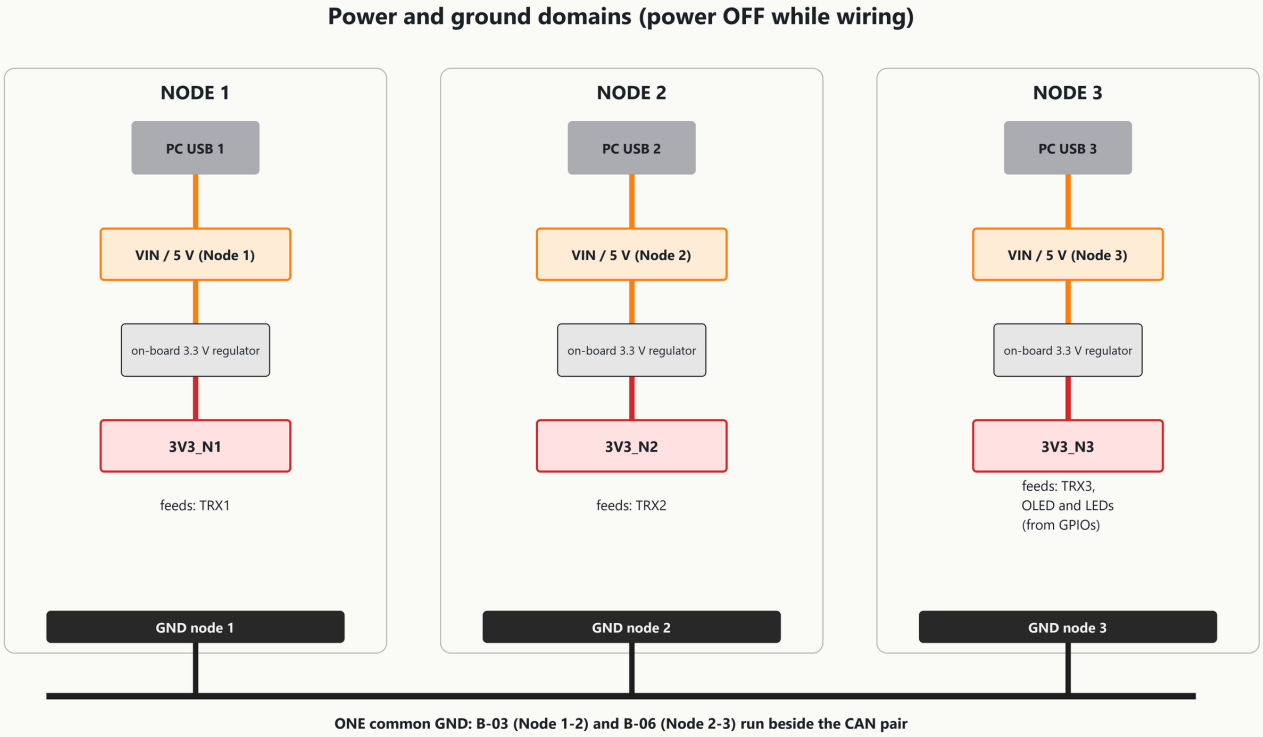
One linear bus: end A - Node 1 - Node 2 - Node 3 - end B. Each node taps the line directly at its transceiver pins; there are no Y branches. Twist each CANH/CANL pair and keep the board-to-board segment short. The GND wire runs beside the pair so all nodes share one reference.

12. Termination: exactly two, at the physical ends

G6 reading per bare module	Action
All three > 10 kOhm (no onboard terminator)	Fit RT1 at TRX1 (N1-05/06) and RT2 at TRX3 (N3-08/09)
TRX1 or TRX3 reads about 120 Ohm	Omit that end's external resistor (the onboard one is the end terminator)
TRX2 reads about 120 Ohm	BLOCKED (row C-T2): its onboard resistor must be removed; send a photo first

ID	From	To	Signal	Wire	Status	Gates	Check (power off)
N1-05	RT1 1	TRX1 CANH	CANH (end A terminator)	lead	**PROVISIONAL**	G6, G13	Fit only if G6 found no onboard terminator on TRX1.
N1-06	RT1 2	TRX1 CANL	CANL (end A terminator)	lead	**PROVISIONAL**	G6, G13	RT1 measured 114-126 Ohm before fitting.
N3-08	RT2 1	TRX3 CANH	CANH (end B terminator)	lead	**PROVISIONAL**	G6, G13	Fit only if G6 found no onboard terminator on TRX3.
N3-09	RT2 2	TRX3 CANL	CANL (end B terminator)	lead	**PROVISIONAL**	G6, G13	RT2 measured 114-126 Ohm before fitting.

13. Ground and power domains



X NEVER join 3V3_N1, 3V3_N2 and 3V3_N3 together (two regulators fighting).

X NEVER join VIN/5 V of different nodes, and NEVER put 5 V on any 3V3 pin or rail.

X No external 5 V supply is used. Nothing is wired to any VIN pin.

X Unplug ALL three USB cables before any wiring change or any resistance / continuity test.

14. Unpowered continuity and resistance checks

All three USB cables unplugged for every check in this section. The PC's USB ground joins the nodes and would falsify readings. Meter leads: black in COM, red in V/Ohm. Never use the current jack.

Check	Probe points	Pass	If not
Ledger walk-through	Each row of section 18, both ends	Every PROVISIONAL row's gates recorded; BLOCKED rows empty	Fix with power off
Supply shorts	3V3 to GND on each node; Node 3 VIN to GND	No steady near-0 Ohm (may drift up as capacitors charge)	Find the short
3V3 isolation	Node 1 3V3 to Node 2 3V3; any VIN to any 3V3	Not near 0 Ohm	A rail or jumper joins them: remove
CANH continuity	TRX1 CANH - TRX2 CANH - TRX3 CANH	About 0 Ohm	Broken trunk
CANL continuity	TRX1 CANL - TRX2 CANL - TRX3 CANL	About 0 Ohm	Broken trunk
Polarity / isolation	CANH or CANL to GND, 3V3, VIN, any ESP32 pin	No continuity	Mis-wire: fix before power
Termination	CANH to CANL anywhere	About 60 Ohm (57-63)	120: one missing; 40: three; open: broken; 0: short
Ground	GND of Node 1, 2, 3 to each other	About 0 Ohm	B-03 / B-06 missing
Strapping pin	D12 (GPIO12)	Nothing inserted	Remove

15. First power-up sequence

- The bus stays isolated from any vehicle or external network.

- Plug in Node 1 only. Serial monitor at 115200 baud: banner lines start with '#'; expect '# CAN init: OK - controller running'. Measure TRX1 3V3-GND: 3.0-3.6 V. Watch for heat, smell, resets. Unplug.

- Repeat for Node 2 (it will report transmit errors alone: no node is powered to ACK; expected).

- Repeat for Node 3. Unplug.

- Unpowered SN65HVD230 nodes do not disturb the bus (TI SLOS346O section 10.3), so the trunk stays wired.

- Never change a wire while any USB cable is plugged in. Unplug at once on heat, smell or repeated resets.

16. One-, two- and three-node bring-up

Stage	Powered	Evidence that counts	Not evidence
1 node	One at a time	Banner shows the expected pins (GPIO17/16), 500 kbit/s, CAN init OK; TRX 3V3 in range	Upload succeeded
2 nodes	Node 1 + Node 2	Node 1 prints JSON lines with "type":"frame" and canId 0x201, 0x202, 0x203 (~1 s) and 0x210 (~0.5 s); error counters stay low; no '# ALERT: BUS-OFF'	Node 2 printing 'Sent CAN ID' (only queued)
3 nodes	all three	Node 3 (monitor, listen-only) prints the same IDs; terminators unchanged	OLED/LED activity (no firmware yet)

The legacy test IDs carry fixed, unverified values (0x201 = 480, 0x202 = 120, 0x203 = 85, 0x210 = 1). They prove frames move; they are not motor or battery signals. Record the actual output in TASK.md and captures/.

Firmware support today

- No firmware drives the OLED, LEDs, buttons or buzzer (GPIO21/22/25/26/27/32/33/18 are not referenced anywhere in firmware/).

- No motor (RPM/temperature) or battery (voltage/current/SoC) simulation and no fault injection exist in the current node1/node2 targets; they only start CAN with TX disabled.

- Only sketch_node2_test (PlatformIO env node2/legacy_sketch) transmits: fixed test frames 0x201, 0x202, 0x203 every 1 s and heartbeat 0x210 every 500 ms, UNVERIFIED units.

- firmware/node3 refuses to start CAN by design. Node 3 hardware (the CANScope monitor) must run the firmware/monitor target, which is listen-only on GPIO17/16 at 500 kbit/s.

17. Common wiring mistakes and troubleshooting

Symptom	Likely cause	Safe response (power off first)
No frames at Node 1	TX/RX swapped (CTX must get TX2); Rs high (standby); CANH/CANL swapped at one node; missing GND; wrong firmware	Check N1-03/04, N2-03/04, G5, polarity
Node 2 errors persist with Node 1 powered	Node 1 not ACKing: Rs standby, RX/TX mis-wired, bit rate mismatch	Compare banners; continuity
Bus reads ~40 Ohm	Third terminator (onboard)	Repeat G6 per module
Bus reads ~120 Ohm	One terminator missing	Check RT1/RT2 or onboard parts
Board will not boot	Something on D12 (strapping) or a short	Remove D12 wire; check rails
LED never lights	Reversed LED, or no firmware drives it yet	Diode test; note firmware gap
OLED blank	No firmware yet; or VCC/GND swapped by position	Re-read labels (G7)
Hot part / smell	Short, reversed supply, 5 V on 3V3	Unplug everything; inspect before reuse

18. Final wire-by-wire sign-off

Fill in on paper with all power off. A BLOCKED row must be physically empty. The electronic version, with coverage columns, is `validation/wire_coverage_checklist.md`.

ID	From -> To	Status	Gates passed	Ends + meter	Init.
N1-01	ESP1 3V3, 3V3-side pin 1 from USB -> TRX1 3V3	PROVISIONAL	[ ]	[ ]	
N1-02	ESP1 GND, 3V3-side pin 2 from USB -> TRX1 GND	PROVISIONAL	[ ]	[ ]	
N1-03	ESP1 TX2 (GPIO17), 3V3-side pin 7 from USB -> TRX1 CTX	PROVISIONAL	[ ]	[ ]	
N1-04	ESP1 RX2 (GPIO16), 3V3-side pin 6 from USB -> TRX1 CRX	PROVISIONAL	[ ]	[ ]	
N1-05	RT1 1 -> TRX1 CANH	PROVISIONAL	[ ]	[ ]	
N1-06	RT1 2 -> TRX1 CANL	PROVISIONAL	[ ]	[ ]	
N2-01	ESP2 3V3, 3V3-side pin 1 from USB -> TRX2 3V3	PROVISIONAL	[ ]	[ ]	
N2-02	ESP2 GND, 3V3-side pin 2 from USB -> TRX2 GND	PROVISIONAL	[ ]	[ ]	
N2-03	ESP2 TX2 (GPIO17), 3V3-side pin 7 from USB -> TRX2 CTX	PROVISIONAL	[ ]	[ ]	
N2-04	ESP2 RX2 (GPIO16), 3V3-side pin 6 from USB -> TRX2 CRX	PROVISIONAL	[ ]	[ ]	
N3-01	ESP3 3V3, 3V3-side pin 1 from USB -> Board B bottom + rail (3V3_N3)	PROVISIONAL	[ ]	[ ]	
N3-02	ESP3 GND, 3V3-side pin 2 from USB -> Board B bottom - rail (GND)	PROVISIONAL	[ ]	[ ]	
N3-03	Board B bottom - rail (GND) -> Board B top - rail (GND)	PROVISIONAL	[ ]	[ ]	
N3-04	Board B bottom + rail (3V3_N3) -> TRX3 3V3	PROVISIONAL	[ ]	[ ]	
N3-05	Board B bottom - rail (GND) -> TRX3 GND	PROVISIONAL	[ ]	[ ]	
N3-06	ESP3 TX2 (GPIO17), 3V3-side pin 7 from USB -> TRX3 CTX	PROVISIONAL	[ ]	[ ]	
N3-07	ESP3 RX2 (GPIO16), 3V3-side pin 6 from USB -> TRX3 CRX	PROVISIONAL	[ ]	[ ]	
N3-08	RT2 1 -> TRX3 CANH	PROVISIONAL	[ ]	[ ]	
N3-09	RT2 2 -> TRX3 CANL	PROVISIONAL	[ ]	[ ]	
N3-10	Board B bottom + rail (3V3_N3) -> OLED VCC	PROVISIONAL	[ ]	[ ]	
N3-11	Board B bottom - rail (GND) -> OLED GND	PROVISIONAL	[ ]	[ ]	
N3-12	ESP3 D21 (GPIO21), 3V3-side pin 11 from USB -> OLED SDA	PROVISIONAL	[ ]	[ ]	
N3-13	ESP3 D22 (GPIO22), 3V3-side pin 14 from USB -> OLED SCL	PROVISIONAL	[ ]	[ ]	
N3-14	ESP3 D25 (GPIO25), VIN-side pin 8 from USB -> R_G 1	BLOCKED	n/a	[ ] empty	
N3-15	R_G 2 -> LED_G A	PROVISIONAL	[ ]	[ ]	
N3-16	LED_G K -> Board B top - rail (GND)	PROVISIONAL	[ ]	[ ]	
N3-17	ESP3 D26 (GPIO26), VIN-side pin 7 from USB -> R_Y 1	PROVISIONAL	[ ]	[ ]	
N3-18	R_Y 2 -> LED_Y A	PROVISIONAL	[ ]	[ ]	
N3-19	LED_Y K -> Board B top - rail (GND)	PROVISIONAL	[ ]	[ ]	
N3-20	ESP3 D27 (GPIO27), VIN-side pin 6 from USB -> R_R 1	PROVISIONAL	[ ]	[ ]	
N3-21	R_R 2 -> LED_R A	PROVISIONAL	[ ]	[ ]	
N3-22	LED_R K -> Board B top - rail (GND)	PROVISIONAL	[ ]	[ ]	
B-01	TRX1 CANH -> TRX2 CANH	PROVISIONAL	[ ]	[ ]	
B-02	TRX1 CANL -> TRX2 CANL	PROVISIONAL	[ ]	[ ]	
B-03	TRX1 GND -> TRX2 GND	PROVISIONAL	[ ]	[ ]	
B-04	TRX2 CANH -> TRX3 CANH	PROVISIONAL	[ ]	[ ]	
B-05	TRX2 CANL -> TRX3 CANL	PROVISIONAL	[ ]	[ ]	
B-06	TRX2 GND -> Board B bottom - rail (GND)	PROVISIONAL	[ ]	[ ]	

Record	Value
ESP32 shield marking N1 / N2 / N3 (G1)	
Transceiver IC marking and labels N1 / N2 / N3 (G4)	
Rs state per module (G5)	
Onboard CANH-CANL reading per module (G6)	
OLED label order and address (G7)	
Bus CANH-CANL resistance, unpowered	
Two-node result (IDs seen at Node 1)	
Three-node result (IDs seen at Node 3)	
Photos saved at	

Record	Value
Checked by / date	

Evidence references

E-TI: Texas Instruments SN65HVD23x datasheet SLOS346O (rev. April 2018): pin 1 D (TXD input), 2 GND, 3 VCC (3.0-3.6 V recommended, 6 V abs max), 4 R (RXD output), 5 Vref, 6 CANL, 7 CANH, 8 Rs. Rs strong pull-down = high-speed, 10-100 kOhm to GND = slope control, >=0.75 VCC = standby (driver OFF, receiver listen-only). D/R abs max VCC + 0.5 V. ICC 10 mA typ / 17 mA max. Package marking VP230.
E-ESP32: Espressif ESP32 Series datasheet: strapping pins GPIO0, GPIO2, GPIO5, MTDI (GPIO12), MTDO (GPIO15); GPIO34-39 input-only with no internal pull-ups; internal pull-up about 45 kOhm.
E-WROOM: Espressif ESP32-WROOM-32 datasheet v3.8: GPIO6-11 connected to the integrated SPI flash (do not use); IO16 (module pin 27) and IO17 (module pin 28) are ordinary GPIOs on WROOM-32 (they are PSRAM pins only on WROVER modules).
E-ARD: framework-arduinoespressif32, variants/esp32/pins_arduino.h (installed in C:/canscope-pio): SDA = 21, SCL = 22 (Wire default pins for board esp32dev); TX = 1, RX = 3 (USB serial UART0).
E-FW: Repository firmware: firmware/node1/include/node1_hw_config.h, firmware/node2/include/node2_hw_config.h and firmware/monitor/include/monitor_hw_config.h set CAN TX GPIO17, RX GPIO16, 500 kbit/s; sketch_node1_test.ino and sketch_node2_test.ino use the same. firmware/node3 refuses to start CAN (all values UNVERIFIED).
E-DOIT: Widely published DOIT ESP32 DevKit V1 30-pin silkscreen order. This is NOT a manufacturer schematic: clone boards vary, so gate G2 compares it with the builder's board.
E-CALC: Hand calculation recorded in validation/correction_register.md (LED current, base drive, termination).